Design, Construction, and Deployment of a Rapidly Developed Emergency Medical Resupply Rover for EVA Support Under Time and Resource Constraints

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***Abstract*—During a crewed Martian surface mission, an astronaut performing an extravehicular activity (EVA) may become injured or immobilized in an area that cannot be immediately reached safely by another crewmember. This work investigates the feasibility of rapidly developing an emergency medical-resupply rover to deliver essential supplies to an isolated astronaut under severe time and resource constraints. The rover was constructed primarily from hardware and materials available within an analog habitat and integrates remote operation, Bluetooth Low Energy (BLE), an Android gateway, MQTT-based communication, telemetry, and a thermally protected medical payload. End-to-end remote control was successfully demonstrated, while preliminary field deployment confirmed the rover's ability to transport the medical payload over outdoor and uneven terrain. Thermal testing showed that the multilayer enclosure reduced heat transfer toward the payload, although further improvement of the thermal-protection system is required. Overall, the study demonstrates the feasibility of rapidly establishing a functional robotic medical-resupply capability using locally available resources during an isolated mission.**

Keywords: emergency medical resupply, planetary rover, extravehicular activity, analog mission, remote operation.

I. Introduction

Future crewed exploration of the Moon and Mars will involve longer-duration, more isolated, and increasingly autonomous surface operations, with limited opportunities for resupply, medical evacuation, and real-time support from Earth [1]. Anticipating operational and medical emergencies before deployment is therefore essential. Analog missions provide a controlled framework for testing systems, procedures, and scenarios relevant to future planetary exploration [2]. In this context, the present work was conducted during a simulated Martian mission at the Analog Astronaut Training Center (AATC) in Poland [3], under isolation, limited resources, and constrained development time.

Surface EVAs are particularly safety-critical because astronauts operate outside the habitat while exposed to demanding terrain, suit-related constraints, and significant physical workloads. NASA reports that future lunar and Martian crewmembers may perform up to 24 hours of EVA per week, increasing the potential for injury or compromised performance [4]. In severe cases, an injured crewmember may become unable to return independently and require continuous assistance, with several such lunar EVA scenarios identified as having potentially catastrophic consequences [5].

Several mitigation concepts have therefore been investigated for an incapacitated EVA crewmember. Walton et al. compared direct crew assistance, walking-assist devices, and wheeled transport, with wheeled systems providing the greatest risk-reduction potential for several severe scenarios [5]. ESA has similarly tested the Lunar Evacuation System Assembly (LESA), a wheeled device designed to recover an incapacitated Moonwalker [6], while NASA's recent Lunar Rescue System Challenge sought compact systems capable of transporting an incapacitated astronaut across difficult lunar terrain [7]. These approaches primarily address evacuation of the injured astronaut. A complementary scenario arises when direct rescue is temporarily unsafe or impossible: emergency medical supplies may instead need to be delivered to the isolated crewmember before evacuation can occur.

This study addresses this scenario through the rapid development of a remotely operated emergency medical-resupply rover during a simulated Martian mission. The system was designed and constructed under limited time and resource availability, relying primarily on components and infrastructure accessible to the crew. Beyond mobility, the design considered remote command and telemetry, operational safety, and preservation of the medical payload during transport, including passive thermal protection.

The main contributions of this work are: (1) the design and deployment of an emergency medical-resupply rover under severe time and resource constraints; (2) the integration of

an end-to-end remote-control and telemetry architecture using readily available communication infrastructure; and (3) the experimental evaluation of a locally constructed thermal-protection solution for the transported medical payload. The study therefore evaluates not only rover mobility, but the feasibility of rapidly establishing a robotic medical-support capability for constrained EVA emergency scenarios.

II. Emergency Scenario and System Requirements

A. Emergency Medical-Delivery Scenario

The considered scenario involves a crewmember performing an EVA on the Martian surface who becomes injured or immobilized in an area that cannot be immediately and safely reached by another astronaut. In such a situation, direct rescue may expose an additional crewmember to unnecessary risk or may simply be temporarily impossible. The proposed rover is therefore intended to provide an intermediate emergency-support capability by transporting essential medical supplies and equipment from the habitat to the isolated crewmember until direct assistance or evacuation becomes possible.

B. System Requirements

The emergency nature of the scenario and the limited resources available during the mission imposed several requirements on the system. The rover had to be rapidly constructed using primarily components and tools already available within the habitat, remotely operated from a safe location, capable of transporting a medical payload over uneven terrain, and able to maintain reliable command and telemetry links during operation. The control architecture also had to provide a safe response to communication loss. Finally, because successful delivery requires the medical payload to remain usable upon arrival, physical and thermal protection of the transported supplies was considered part of the overall system design.

III. Rover Design and Implementation

A. System Overview

Figure 1 presents the overall architecture of the proposed emergency medical-resupply system. The left-hand side shows the end-to-end command and data flow between the remote control station, cloud services, smartphone gateway, and rover-side electronics. The right-hand side provides an expanded view of the onboard electronic chain responsible for wireless command reception, motor control, and actuation.

The rover-side unit integrates the smartphone and embedded electronics. Commands generated at the control station are transmitted through cloud services to the smartphone, which acts as a gateway to the rover electronics. In the opposite direction, rover telemetry, positioning information, and video data can be returned to the remote operator. This architecture prioritizes readily available infrastructure and reduces the need for dedicated long-range communication and sensing hardware.

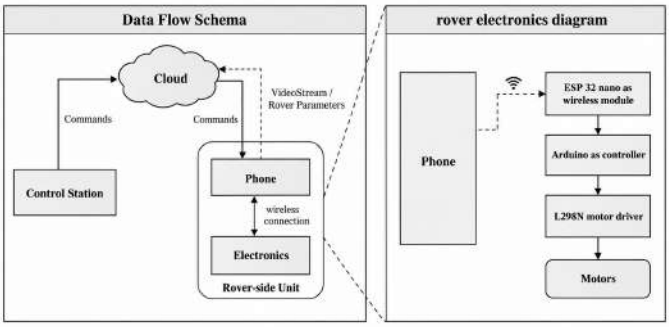


Fig. 1. Overall rover architecture. Left: data-flow schema showing command and telemetry exchange between the control station, cloud, and rover-side unit. Right: exploded view of the rover electronic architecture, detailing the smartphone gateway, wireless module, controller stage, motor driver, and motors.

B. Mechanical Design and Rapid Prototyping

No dedicated rover platform or complete set of components was prepared before the mission. The mechanical system was therefore developed primarily from hardware available within the habitat, with reuse and limited local fabrication favored to reduce development time.

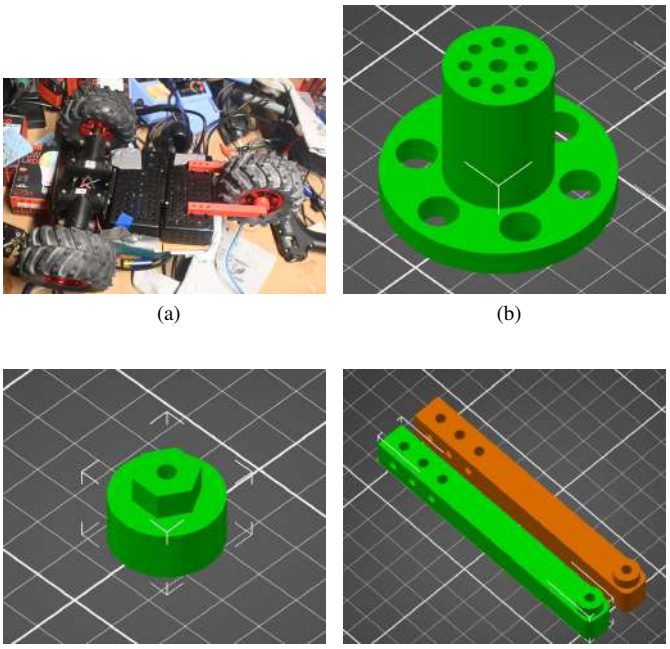
An incomplete metallic rover chassis was selected as the mechanical basis of the platform. The final configuration consists of three wheels, including two 75:1 DC geared drive motors coupled to large wheels and an additional non-driven wheel integrated into the existing structure. The geared drivetrain was selected to favor low-speed torque during traversal of uneven terrain.

Because the additional wheel had no compatible mounting interface, dedicated mechanical adapters were designed and produced by 3D printing. These parts enabled integration of the wheel while preserving most of the original chassis. Figure 2 summarizes the resulting platform and the principal custom components.

C. Electronic Control and Safety

The final onboard control chain consists of an ESP32 Nano used as the wireless interface, an Arduino controller, an L298N dual H-bridge motor driver, and the two DC drive motors, as illustrated in Fig. 1. Commands received through BLE are transferred to the control stage and converted into independent left- and right-motor actuation, enabling differential-drive motion.

The communication protocol uses compact command packets containing a sequence number, command identifier, payload, and checksum. Rover telemetry provides status information and command acknowledgments to the remote interface. A communication watchdog automatically stops the drive motors when valid commands are no longer received within the configured timeout, providing a fail-safe response to communication loss.



(a) (b) (c) (d)

Fig. 2. Mechanical implementation of the improvised rover: (a) final three-wheel platform based on a repurposed chassis; (b)–(d) custom components designed and 3D-printed to integrate the additional wheel with the existing structure.

D. Embedded Control and Safety

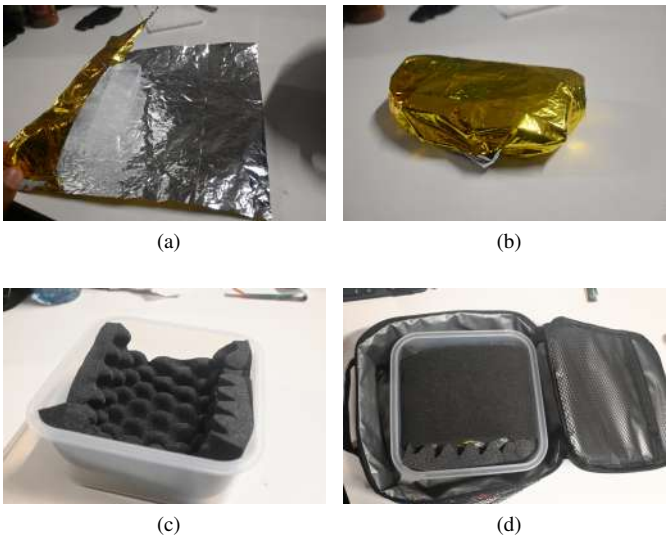
The rover uses a differential-drive locomotion scheme in which steering and forward motion are achieved through independent control of the left and right motors. The controller interprets incoming commands and translates them into motor actuation values. In addition to motion control, the embedded logic supports command verification, telemetry exchange, and safety handling.

To reduce the risk of uncontrolled motion, a watchdog mechanism was included so that the rover automatically stops when valid commands are no longer received within a specified interval. This fail-safe behavior was considered essential for an emergency-support rover intended to operate in constrained environments.

E. Communication and Remote Operation

The rover supports both local and remote operation. For local control, the Android application provides a joystick interface that communicates directly with the rover through Bluetooth Low Energy (BLE). For remote operation, commands generated from either a desktop control interface or a Telegram bot are transmitted through an MQTT broker to the Android gateway. The gateway then forwards the commands to the rover over BLE.

The communication path is bidirectional. Rover telemetry received by the Android gateway through BLE is republished through MQTT, allowing remote interfaces to display rover status and operational parameters. The Android application



(a) (b) (c) (d)

Fig. 3. Successive layers of the medical payload protection system: (a) inner medication container and emergency blanket; (b) wrapped inner payload; (c) foam-lined rigid enclosure; and (d) assembled payload placed inside the insulated transport bag.

therefore acts as the central bridge between the rover's local BLE link and the remote communication infrastructure. In addition, the smartphone provides onboard sensing capabilities, including positioning and inertial measurements, which can be incorporated into the transmitted telemetry.

To improve communication robustness, the gateway includes automatic BLE scanning and reconnection mechanisms, while the MQTT communication layer supports automatic reconnection after network interruptions. This architecture allowed both direct local control and remote supervision without requiring a dedicated long-range radio system on the rover.

F. Medical Payload and Thermal Protection

The medical payload was treated as an integral subsystem of the rover, since successful delivery requires the transported supplies to remain physically protected and within acceptable environmental conditions. A passive multilayer enclosure was therefore assembled using materials available within the habitat.

The medication was first placed inside a compartmentalized plastic container and wrapped with an emergency blanket to reduce radiative heat exchange. Bubble wrap was subsequently added to introduce an additional trapped-air insulation layer. The protected payload was then placed inside a rigid food-storage container lined with foam, providing additional thermal resistance and mechanical protection. Finally, the complete assembly was placed inside an insulated transport bag to provide an additional environmental barrier during rover deployment.

Figure 3 illustrates the successive layers of the payload-protection system. The thermal performance of this configuration was evaluated separately through a controlled thermal-stress experiment described in Section IV-C.

IV. Experimental Methodology

A. End-to-End Control Validation

The integrated command chain was evaluated by transmitting motion commands from the remote control interface through the communication architecture to the rover. Commands were relayed through the Android gateway and BLE link to the onboard controller, where they were translated into left- and right-motor actuation. Successful command execution was verified from the observed motor response and from telemetry returned through the same communication chain.

The test also verified bidirectional communication between the rover and the remote interface, including command acknowledgments and rover-status updates. Communication-loss handling was assessed through the embedded watchdog, which was configured to stop motor actuation when valid commands were no longer received within the specified timeout.

B. Field Mobility and Emergency-Delivery Scenario

A field scenario was defined to evaluate the rover beyond bench-level operation under conditions representative of an emergency medical-delivery task. The selected route remained close to the habitat to limit recovery complexity while including outdoor terrain, local road segments, and predefined control points, as shown in Fig. 4a.

During deployment, the rover was remotely operated from the control station while carrying the insulated medical payload. Mobility was evaluated over natural grass terrain and a modular uneven surface introduced to assess the ability of the three-wheel platform to maintain motion over local geometric irregularities. Figure 4 summarizes the field-test configuration and rover deployment conditions.

C. Thermal Stress Evaluation

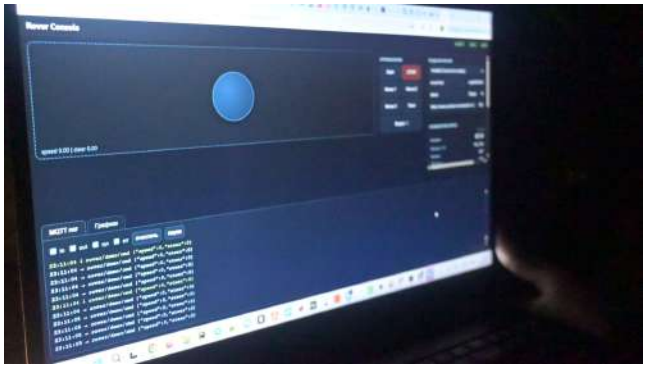
The thermal-protection subsystem was evaluated using an accelerated conductive thermal-stress test. The insulated medical container, containing ibuprofen (400 mg) as a representative payload, was placed on the heated bed of a 3D printer. Before heating, the temperature of the medical sample and the outer surface of the rigid enclosure were measured using a thermal camera.

The printer bed was maintained at 60°C for the first 35 min and then increased to 80°C for the remaining exposure period, resulting in a total test duration of 60 min. At the end of the experiment, the temperature of the outer enclosure was measured again. The payload was then removed from the insulated assembly and its surface temperature was measured immediately after unpacking.

The experiment was intended to characterize the attenuation provided by the improvised multilayer insulation under a strong imposed heat load. It should therefore be interpreted as an accelerated conductive thermal-stress test rather than as a direct reproduction of the Martian thermal environment.



(a) Field route and control points.



(b) Remote control station during rover operation.



(c) Traversal of the modular uneven surface.

Fig. 4. Field evaluation of the emergency medical-resupply rover: (a) planned traverse route with predefined control points; (b) remote operator control interface; and (c) mobility test over a modular uneven surface.

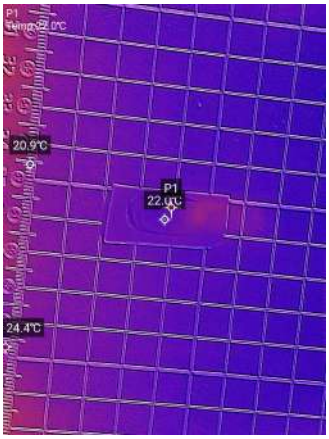
TABLE I
Thermal response of the medical payload enclosure after a 60-min thermal-stress exposure.

Measurement	Initial	Final	ΔT
Outer enclosure	23.5°C	50.0°C	26.5°C
Medical sample	21.7°C	32.0°C	10.3°C

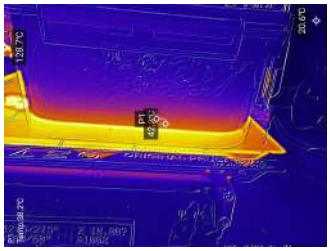
V. Results

A. System Integration and Remote Control

The complete control architecture was successfully integrated from the operator interface to the rover propulsion system. Commands transmitted through the Android gateway and BLE link produced the expected left- and right-motor responses, while rover telemetry was received by the mobile



(a) Initial thermal condition of the representative medical sample.



(b) Insulated enclosure during the thermal-stress exposure.

Fig. 5. Thermal-stress experiment: (a) initial thermal condition of the representative medical payload; and (b) insulated enclosure exposed to the heated 3D-printer bed.

interface. Remote command transmission through the MQTT-based communication path was also demonstrated, confirming functional end-to-end integration of the communication, control, actuation, and telemetry subsystems.

B. Field Deployment

The rover was deployed outdoors while carrying the insulated medical payload. The platform demonstrated controlled motion over natural grass and the modular uneven surface shown in Fig. 4, confirming that the three-wheel configuration could transport the payload beyond bench-level testing.

The complete planned traverse shown in Fig. 4a was not performed. Field testing was limited by the remaining mission time and by the condition of the battery recovered from the habitat. The available battery was already significantly degraded and provided insufficient current and endurance for sustained rover operation. Consequently, the field experiment demonstrates rover mobility and medical-payload transportation, but does not constitute a complete validation of the proposed emergency delivery route.

C. Thermal Protection Performance

The thermal measurements reported in Table I show that the representative medical sample experienced a substantially smaller temperature increase than the outer enclosure. During the 60-min thermal exposure, the outer enclosure temperature increased by 26.5°C, whereas the medical sample increased by 10.3°C. This difference indicates that the multilayer passive enclosure provided measurable thermal attenuation.

The final sample temperature reached 32°C, exceeding the 25°C storage limit specified for several marketed ibuprofen 400-mg tablet formulations [8]–[10]. However, because the exact commercial formulation used in the experiment was not characterized and no pharmaceutical stability or potency analysis was performed, this result should not be interpreted as evidence of drug degradation. The experiment therefore demonstrates the thermal response of the improvised enclosure rather than pharmaceutical preservation.

VI. Discussion

The results demonstrate the feasibility of rapidly assembling and operating an emergency medical-resupply rover using primarily hardware, materials, and infrastructure already available within an isolated habitat. A major engineering lesson was the advantage of simplicity under severe time constraints. Earlier attempts to reuse more complex motor-control hardware required substantial troubleshooting, including logic-level incompatibility, hardware faults, and power-supply issues. Replacing this architecture with a simpler and better-understood control chain substantially reduced integration time. The same principle influenced the communication architecture: an existing smartphone, BLE, MQTT services, and readily available software infrastructure were favored over dedicated long-range communication and telemetry hardware. These choices introduced limitations, but enabled a functional end-to-end system to be produced within the mission duration.

The present study remains a preliminary feasibility demonstration. The complete emergency-delivery route could not be executed because of limited mission time and the degraded battery available in the habitat, while communication latency, packet loss, energy consumption, and route-completion time were not quantitatively characterized. The thermal experiment should also be interpreted as an accelerated conductive stress test rather than a direct reproduction of Martian thermal conditions. Future evaluation should therefore use a properly characterized power system, repeat the complete delivery route over multiple trials, record command reliability, latency, current consumption, and travel time, and continuously monitor payload temperature using an internal sensor. Further developments could also investigate improved thermal protection, communication systems independent of terrestrial cellular infrastructure, obstacle detection, and semi-autonomous navigation.

VII. Conclusion and Future Work

This work presented the design, construction, and preliminary deployment of a rapidly developed emergency medical-resupply rover intended to support EVA operations under severe time and resource constraints. The resulting system integrated a repurposed three-wheel mobile platform, remote control, bidirectional telemetry, smartphone-based communication, and a protected medical payload. End-to-end control was successfully demonstrated, and the rover achieved outdoor mobility while transporting the medical container. The passive payload enclosure also reduced the thermal load transmitted to the representative medication, although the test did not demonstrate that the payload remained within the selected reference storage range.

The study therefore supports the feasibility of creating a robotic medical-resupply capability from locally available resources during an isolated mission, while also identifying the limitations that must be addressed before such a concept can be

considered operationally mature. Future work should focus on full-route validation with a reliable power system, quantitative communication and energy measurements, improved payload thermal management, repeated field trials, and progressively greater navigation autonomy.

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